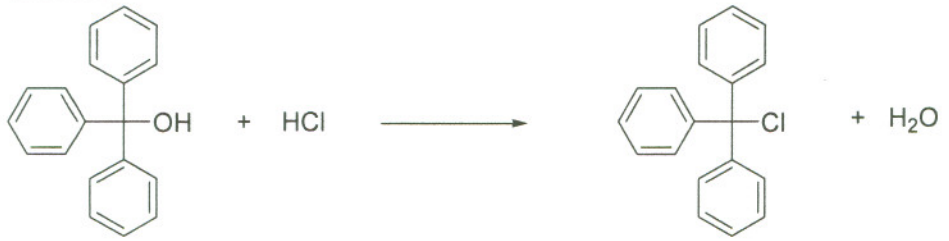
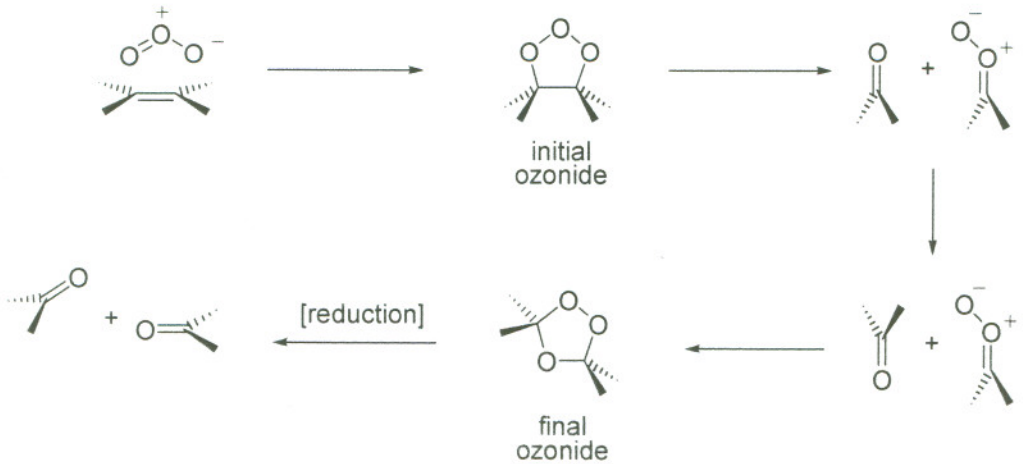
Organic Cume 1/21/2011

1. Give the HMO molecular orbital relative energies and the coefficient signs (math signs) for allene, 1,2-propadiene. (This question is not asking for a calculation, this should be a qualitative energy diagram of the MO's for the allyl anion with coefficient math signs).

2. Consider an E2 elimination reaction between 2-bromobutane and potassium tert-butoxide ($KOC(CH_3)_3$) to give 2-butene. If you wanted to investigate this reaction mechanistically with kinetic isotope studies what would you propose to study? What would you expect to observe?

3. For the reaction shown below answer the following questions. (a) Write the mechanisms for the reactions. (b) Indicate whether a Hammett plot for <u>rates</u> of this reaction should have a positive or negative slope. It is important to explain this part of your answer.



4. Are the following conversions are allowed or forbidden? Explain using any orbital symmetry approach that you like.

5. In the mechanism for the ozonolysis of ethene is shown below. Show the orbital symmetry that predict for the cycloaddition reaction that forms the initial ozonide.